

COLLEGE FOOTBALL FANTASY

AI/ML Requirements & Use Cases

MVP-Focused Revision — TRD & Backlog

Project	College Football Fantasy — Team 4
Sprint	Sprint 1 — Foundation
Owner Function	AI/ML
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1. Purpose

Define the practical AI/ML requirements needed to support the College Football Fantasy MVP. The AI/ML layer shall improve player decision-making through performance predictions, lineup recommendations and player insights while remaining separate from authoritative scoring, rankings and game rules.

2. AI/ML MVP Scope

MVP — Mandatory

- Player performance prediction.
- Optimal lineup recommendations.
- Player insights based on reliable player and game data.
- Defined player/game data requirements.
- Defined model inputs, outputs and evaluation approach.
- Secure integration between the AI service, backend and application.
- Monitoring of data quality and model performance.

Principle

- AI outputs are recommendations/predictions; the backend remains authoritative for scores, rankings, roster rules and game results.

3. Player Performance Prediction

MVP — Mandatory

- Predict expected fantasy performance or expected fantasy points for eligible players.
- Use historical and current player/game statistics available to the platform.
- Account for relevant contextual information such as opponent/team and recent form where data quality permits.
- Return a prediction with sufficient metadata to identify the model/version and prediction period.
- Present predictions as estimates rather than guaranteed outcomes.
- Establish a simple non-ML baseline for comparison.

Future Enhancement

- More advanced forecasting architectures, deeper contextual modelling and highly personalized predictions.

4. Optimal Lineup Recommendations

MVP — Mandatory

- Recommend players for a user's fantasy lineup using model predictions and applicable roster rules.
- Respect position, roster, eligibility and gameweek constraints defined by the product.
- Use predicted player performance as an input rather than allowing the model to bypass product rules.
- Provide recommendations through a backend API.
- Clearly distinguish recommended actions from actions actually executed by the user.
- Prevent AI recommendations from directly changing a user's lineup without the normal authorized product flow.

Future Enhancement

- Advanced automated lineup optimization and deeper scenario-based optimization.

5. Player Insights

MVP — Mandatory

- Provide concise insights derived from available player statistics and model outputs.

- Highlight useful trends such as recent form, consistency and relevant matchup/context signals where supported by data.
- Avoid unsupported claims or fabricated statistics.
- Clearly identify data-driven insights as informational.
- Ensure insights use current or appropriately timestamped data.

Future Enhancement

- Advanced personalized insights, natural-language explanations and deeper player comparisons.

6. Required Player & Game Data

MVP — Required Data

- Player identifier and position.
- Team/college information.
- Historical game performance.
- Recent performance/form statistics.
- Game and gameweek information.
- Opponent information.
- Fantasy scoring data and applicable scoring rules.
- Player eligibility/status data where reliably available.
- Data source and timestamp/freshness information.

Data Quality Requirements

- Validate incoming data before model use.
- Detect missing, stale, duplicated or inconsistent records.
- Maintain traceability to the source of important model inputs.
- Define how unavailable data is handled rather than silently inventing values.

Future Enhancement

- Additional contextual, environmental and advanced performance datasets.

7. Model Inputs, Outputs & Evaluation

MVP — Inputs

- Historical player statistics.
- Recent performance indicators.
- Game/opponent context.
- Player/team attributes required by the selected model.
- Relevant fantasy scoring features.

MVP — Outputs

- Expected player performance/fantasy points.
- Confidence or uncertainty indicator where technically appropriate.
- Lineup recommendation score/ranking.
- Player insight signals.

MVP — Evaluation

- Use appropriate regression metrics such as MAE and/or RMSE for prediction accuracy.
- Evaluate lineup recommendations against defined recommendation-quality measures.

- Compare the ML model with a simple baseline.
- Evaluate performance on data separated from training data.
- Track performance over time after deployment.

Future Enhancement

- More advanced calibration, explainability and model experimentation frameworks.

8. AI/ML Architecture & Backend/App Integration

MVP — Mandatory

- AI inference should be performed by a trusted backend/service layer rather than directly in the client.
- The application should request predictions, insights and recommendations through authenticated backend APIs.
- The backend should validate user authorization before returning user-specific AI recommendations.
- AI services must not directly modify authoritative scores, rankings, roster rules or game results.
- AI outputs should be treated as recommendations/predictions by the product.
- API failures or unavailable model predictions must fail safely without preventing core fantasy functionality where possible.
- AI endpoints must follow the project's API security, authentication, authorization and rate-limiting requirements.

Logical Flow

- Player/Game Data → Data Validation → ML Model → Prediction/Insight → Backend API → Application UI → User Decision.

9. Monitoring & Data Quality

MVP — Mandatory

- Monitor missing, stale, duplicated and inconsistent input data.
- Monitor model prediction performance against actual outcomes.
- Track model/version information for important predictions.
- Record relevant data-quality and model-performance events.
- Establish thresholds or review triggers for significant degradation.
- Provide a safe fallback when model inputs are unavailable or invalid.

Future Enhancement

- Automated data-drift detection, model-drift detection and advanced ML observability dashboards.

10. AI/ML Security & Integrity

MVP — Mandatory

- AI endpoints must require appropriate authentication and authorization.
- Model inputs must be validated server-side.
- Users must not be able to manipulate authoritative game data through AI endpoints.
- AI outputs must not override backend scoring or ranking logic.
- Model files, API keys and service credentials must be securely stored.
- Sensitive user data must not be unnecessarily exposed to the model.
- AI-related logs must not expose secrets or unnecessary personal information.

11. MVP Use Cases

UC-01 — Predict Player Performance

- User views an eligible player.

- Backend retrieves approved player/game data.
- AI service generates expected performance.
- Backend returns the prediction and relevant insight.
- User uses the prediction as decision support.

UC-02 — Recommend Optimal Lineup

- User requests lineup assistance.
- Backend verifies the user's fantasy team and applicable rules.
- AI service evaluates eligible players.
- Recommendation is returned.
- User accepts, rejects or adjusts the recommendation through the normal lineup workflow.

UC-03 — View Player Insights

- User requests player insights.
- Backend supplies validated current/recent data.
- AI/model logic generates supported insights.
- Application displays the insights with appropriate context.

UC-04 — Handle Poor Data Quality

- Required data is missing, stale or invalid.
- Data-quality control prevents unreliable model input.
- System returns an appropriate fallback or unavailable-state message.

12. Future AI/ML Features

These should remain in the TRD/backlog and should not be required for the MVP:

- Trade recommendations.
- Waiver recommendations.
- Injury prediction.
- Advanced fraud/cheating detection using ML.
- Advanced automated lineup optimization.
- Advanced personalization.
- More sophisticated player forecasting.
- Automated anomaly detection.
- Advanced behavioral modelling.

13. MVP vs Future Feature Summary

MVP — Player performance prediction; optimal lineup recommendations; player insights; required player/game data; model inputs/outputs/evaluation; backend/app integration; monitoring and data-quality controls.

Future — Trade/waiver suggestions; injury prediction; advanced fraud detection; advanced personalization; advanced anomaly detection; more sophisticated forecasting.

14. Acceptance Criteria

- MVP can generate a player performance prediction from validated data.
- Prediction quality is evaluated against a defined baseline.
- MVP can produce lineup recommendations while respecting roster/gameweek rules.

- Player insights are based on available validated data.
- Required data inputs and their sources are documented.
- AI outputs, inputs and evaluation approach are documented.
- AI functionality is accessed through secured backend services.
- AI cannot directly alter authoritative scores, rankings or game rules.
- Data-quality failures are detected and handled safely.
- Model performance can be monitored after deployment.
- Future AI features are clearly separated from MVP scope.

15. Conclusion

The College Football Fantasy MVP should use AI/ML as a focused decision-support capability. The initial implementation should concentrate on player performance prediction, lineup recommendations and player insights, supported by reliable data, measurable model evaluation, secure backend integration and practical monitoring. Advanced features remain future enhancements and should be tracked in the TRD/backlog.